Master Guide: Installing RMFIT on Kubuntu 24.04+

Documentation

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Overview

This guide details the procedure for installing the legacy **RMFIT** software on modern Linux distributions (Kubuntu/Ubuntu 24.04 and later). It addresses critical issues regarding missing legacy libraries (specifically `libXp`) and version compatibility.

1 Phase 1: Preparation (Host Side)

Before working in the terminal, download the source file correctly.

1. Go to the Fermi SSC Legacy Page.
2. Locate the link for **"The Linux Distribution (64 bit version)"** (filename: `rmfit_v432_64bit.tar.gz`).
3. **Important:** Right-click and choose *"Save Link As..."* to ensure the file downloads completely (approx. 100MB).

2 Phase 2: System Dependencies

Run the following commands in your terminal to install the necessary build tools and legacy compatibility layers.

1. Update and install standard tools

```
sudo apt update
sudo apt install -y tcsh wget make binutils xfonts-base xfonts-100dpi
    xfonts-75dpi
```

2. Install standard X11 libraries

These libraries are available in the official repositories and are required for the RMFIT GUI.

```
sudo apt install -y libxmu6 libxpm4 libxt6 libice6 libsm6 libxext6
    libx11-6
```

3. The "Ncurses" Compatibility Fix

RMFIT requires Ncurses version 5, but Ubuntu 24.04 ships with version 6. We create symbolic links to trick the software.

```
sudo apt install -y libncurses6 libtinfo6
sudo ln -s /usr/lib/x86_64-linux-gnu/libncurses.so.6 /usr/lib/x86_64-
   linux-gnu/libncurses.so.5
sudo ln -s /usr/lib/x86_64-linux-gnu/libtinfo.so.6 /usr/lib/x86_64-linux
   -gnu/libtinfo.so.5
```

3 Phase 3: The "Missing" Library Fix (libXp)

The `libXp` library was removed from Ubuntu years ago. We must manually download and place it in the system folder.

1. Download the old package

```
mkdir -p ~/temp_libxp && cd ~/temp_libxp
wget http://launchpadlibrarian.net/183708483/libxp6_1.0.2-2_amd64.deb
```

2. Extract the library file

Note: Do not use `apt install` or `dpkg -i`. We are extracting the file manually to avoid dependency errors.

```
ar x libxp6_1.0.2-2_amd64.deb
tar -xf data.tar.xz
```

3. Install and Cleanup

Copy the library to the system path and remove temporary files.

```
sudo cp usr/lib/x86_64-linux-gnu/libXp.so.6.2.0 /usr/lib/x86_64-linux-
   gnu/libXp.so.6
cd ~
rm -rf ~/temp_libxp
```

4 Phase 4: Install RMFIT

We install the software to `/opt` to keep it isolated from personal files.

1. Prepare the directory

```
sudo mkdir -p /opt/rmfit
sudo chown $USER /opt/rmfit
```

2. Copy and Extract

Replace the source path with the actual location of your download.

```
cp ~/Downloads/rmfit_v432_64bit.tar.gz /opt/rmfit/
cd /opt/rmfit
# Use -xvf (not -zxvf) to avoid "not in gzip format" errors
tar -xvf rmfit_v432_64bit.tar.gz
```

5 Phase 5: Run RMFIT

To launch the software:

```
cd /opt/rmfit/rmfit_v432
./rmfit_v432
```

Expected Result: The IDL Virtual Machine window will appear. Click the splash screen to enter the program.

Troubleshooting

- **Error: "gzip: stdin: not in gzip format"**: This occurs if the browser decompressed the file during download. Use `tar -xvf` instead of `tar -zxvf`.
- **Error: "libpng12.so.0 not found"**: If this error appears, repeat the *Phase 3* procedure using an old `libpng12` deb file.
- **Small/Ugly Fonts**: This is normal for legacy IDL applications on high-resolution displays. It relies on the `xfonts` packages installed in Phase 2.